

Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)

Installed docker in EC2

```
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-7-194:~$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
ubuntu@ip-172-31-7-194:~$
```

i-043f0e277c8163d51 (Docker Machine)

PublicIPs: 3.110.44.143 PrivateIPs: 172.31.7.194

Created the Docker images

```
---> 0a903cd17b82
Successfully built 0a903cd17b82
Successfully tagged taskimage:v1
root@ip-172-31-7-194:/home/ubuntu# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
taskimage      v1        0a903cd17b82   About a minute ago  555MB
ubuntu         latest    c3a134f2ace4   5 weeks ago    78.1MB
root@ip-172-31-7-194:/home/ubuntu#
```

i-043f0e277c8163d51 (Docker Machine)

PublicIPs: 3.110.44.143 PrivateIPs: 172.31.7.194

Created the Docker Containers

```
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
taskimage      v1        0a903cd17b82   About a minute ago  555MB
ubuntu         latest    c3a134f2ace4   5 weeks ago    78.1MB
root@ip-172-31-7-194:/home/ubuntu# docker run -d -p 8080:80 taskimage:v1
6339d57044505051dac2c2651de25d54fae7d9c930285d5b28a281cbef3010e5
root@ip-172-31-7-194:/home/ubuntu# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS      PORTS          NAMES
6339d5704450   taskimage:v1   "python3 app.py"        About a minute ago  Exited (0)  About a minute ago  unruffled_mendeleev
```

```
Run 'docker logs --help' for more information
root@ip-172-31-7-194:/home/ubuntu# docker logs 6339d5704450
Hello, Docker!
root@ip-172-31-7-194:/home/ubuntu#
```

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Created the Docker Volumes

```
Run 'docker --help' for more information
root@ip-172-31-7-194:/var/lib/docker/volumes/data# docker volume ls
DRIVER      VOLUME NAME
local       data
root@ip-172-31-7-194:/var/lib/docker/volumes/data#
```

i-043f0e277c8163d51 (Docker Machine)

PublicIPs: 3.110.44.143 PrivateIPs: 172.31.7.194

Docker network

```
root@ip-172-31-7-194:/var/lib/docker/volumes/data# docker network ls
NETWORK ID      NAME      DRIVER      SCOPE
c5aa53959730    bridge    bridge      local
7689162762c7    host      host        local
6bd3f948bf9c    none      null        local
root@ip-172-31-7-194:/var/lib/docker/volumes/data#
```

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